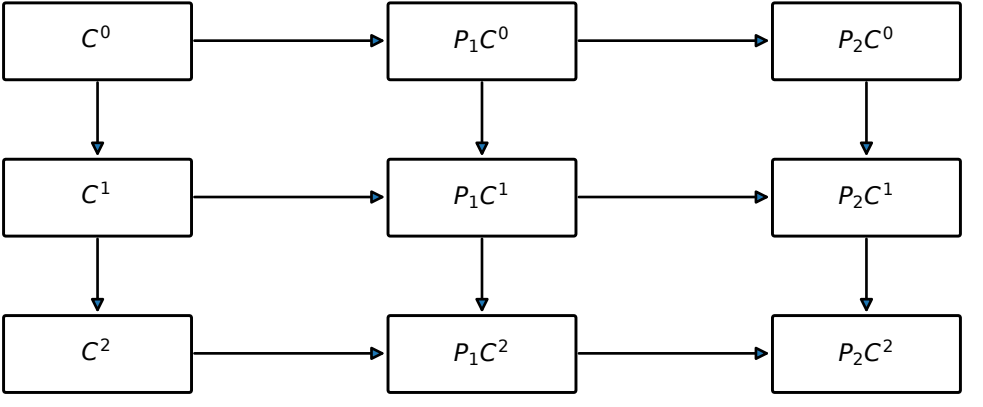
chain projectors split the full correction complex



$$H^r(C^\bullet) \cong H^r(P_1C^\bullet) \oplus H^r(P_2C^\bullet)$$